Year 13 Maths - Moments

Similar questions - Beams on ropes, hinged rods and laminas

Two new questions on this topic, a little more open-ended than the exam questions. They are not past exam questions. Answers are at the end.

1 A beam AB of length 6 m and mass 30 kg is suspended horizontally by two vertical light inextensible ropes, one attached at C where $AC = 1$ m and the other at D where $DB = 0.8$ m. The beam is modelled as a non-uniform rod. The tension in the rope at C is twice the tension in the rope at D. Take $g = 9.8$.

(a) Find the distance of the centre of mass of the beam from A. **[5]**

(b) A particle of mass m kg is attached to the beam at A. The beam remains in equilibrium. Find the largest possible value of m. **[3]**

2 A uniform horizontal rod AB of length 1 m and mass 3 kg is freely hinged to a vertical wall at A. A particle of mass 2 kg is attached at B. The rod is held in equilibrium by a light inextensible string joining the midpoint C of the rod to a point D on the wall, 0.6 m vertically above A. Take $g = 9.8$.

(a) Find the tension in the string. **[4]**

(b) Find the magnitude and direction of the force exerted on the rod at A. **[5]**

3 A uniform rectangular lamina $ABCD$ has mass 3 kg, $AB = 1$ m and $AD = 4$ m. The lamina rests in a vertical plane with A on rough horizontal ground and D against a smooth vertical wall, with AD inclined at 65° to the horizontal and B on the side of AD away from the wall. Take $g = 9.8$.

(a) By taking moments about A, find the normal contact force between the lamina and the wall. **[4]**

(b) The lamina is in limiting equilibrium. Find the coefficient of friction between the lamina and the ground. **[3]**

Question 1

(a) Total tension $= 30g$, so $T_D = 10g$ and $T_C = 20g$. Moments about A (x is the centre of mass distance): $20g \times 1 + 10g \times 5.2 = 30g\,x$, so $72 = 30x$ and $x = 2.4$ m

(b) The beam is about to tip about C when $T_D = 0$. Moments about C: $mg \times 1 = 30g \times (2.4 - 1) = 42g$, so $m \leqslant 42$. Largest $m = 42$ kg

Question 2

(a) The string makes angle ϕ with the horizontal where $\tan\phi = \dfrac{0.6}{0.5}$, so $\sin\phi = \dfrac{0.6}{\sqrt{0.61}} = 0.768$ and $\cos\phi = 0.640$.

Moments about A: $T\sin\phi \times 0.5 = 3g \times 0.5 + 2g \times 1 = 3.5g$, so $T = \dfrac{7g}{\sin\phi} = 89.3$ N

(b) Horizontal component at A: $X = T\cos\phi = 57.2$ N (away from the wall). Vertical: $T\sin\phi = 7g = 68.6$ and weights total $5g = 49$, so A must supply 19.6 N vertically downwards. Magnitude $\sqrt{57.2^2 + 19.6^2} = 60.4$ N at 18.9° below the horizontal

Question 3

(a) The centre of mass is at the centre of the lamina. Taking A as origin, with $\overrightarrow{AD}$ of length 4 at 65° towards the wall, G is 2 along AD and 0.5 along AB (perpendicular). Horizontally G is $2\cos 65^\circ - 0.5\sin 65^\circ = 0.392$ m on the wall side of A. D is $4\sin 65^\circ = 3.625$ m above A. Moments about A: $S \times 3.625 = 3g \times 0.392$, so $S = 3.18$ N

(b) Horizontally $F = S = 3.18$ N and vertically $N = 3g = 29.4$ N. $\mu = \dfrac{F}{N} = 0.108$